

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	RENEE VINCENT ASR	Reg. No.:	192424161
Section / Batch:	BTECH AI & DS	Date:	17-08-2026

Code of Conduct

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO3 - AT3
MCQ TEST

RENEE VINCENT ASP
192424161
CO3-AT3-MCQ

37
50

- 1) D) ✓
- 2) D) Lexical Ambiguity →
- 3) B) Parse Trees →
- 4) B) Subcategorization frame →
- 5) B) →
- 6) A) ✓
- 7) C) →
- 8) A) ✓
- 9) B) Early parser ✓
- 10) A) ✓
- 11) B) Add recursive →
- 12) B) →
- 13) C) ✓
- 14) B) Left recursion ✗
- 15) D) Dynamic chart →
- 16) B) ✓
- 17) A) Tree 2 because it is shorter ✗
- 18) C) ✓
- 19) B) ✓
- 20) C) ✓
- 21) C) ✓
- 22) B) ✓
- 23) ~~B)~~ ✓
- 24) B) →

25) C) ✓

26) B) ✗

27) B) ✗

28) B) ✓

29) B) Early Parser ✓

30) B) Highest PCFG rule ✓

31) B) ✗

32) A) ✓

33) A) ✓

34) B) ✗

35) A) Predictor ✓

36) B) ✗

37) A) ✓

38) A) Early Parsing ✓

39) B) ✗

40) A) ✓

41) C) $S \rightarrow NP VP$ ✓

42) B) VP ✓

43) B) ✓

44) A) ✗

- 44) b) features automatically converts to plural. ✓
- 45) d) the highest product of rule probabilities. ✓
- 46) a) CFG supports recursive noun phrase ✓
- 47) a) Morphological Ambiguity. ✗
- 48) c) recursive grammar expansion. ✗
- 49) a) The verb "gaze" requires additional complements. ✓
- 50) c) V NP PP ✓